Apache OpenOffice

Version 4.1

Impress Guide

AOO Documentation Team

Chapter 6
Formatting Graphic Objects

Acknowledgments

This book is updated from a previous version by the OOoAuthors team. The AOO Documentation Team appreciates the many contributions of Michael Scarborough and Michele Petrovsky.

Publication date and software version

Published October 30, 2025. Based on Apache OpenOffice 4.1.

Note for Mac Users

Some keystrokes and menu items differ on a Mac from those used in Windows and Linux. The table below gives some common substitutions for the instructions in this chapter. For a more detailed list, see the application Help.

Windows/Linux	***Mac equivalent***	***Effect***
Tools > Options menu selection	**OpenOffice > Preferences**	Access setup options
Right-click	*Control+click*	Open context menu
Ctrl (Control)	⌘ (Command)	Used with other keys
F5	*Shift+⌘+F5*	Open the Navigator
F11	*⌘+T*	Open Styles & Formatting window

Contents

Formatting Objects

This chapter describes how to format graphic objects created with available drawing tools.

The format of each graphic object, in addition to its size, rotation, and position on the slide, is determined by several attributes that define the line, text, and area fill of each object. These attributes, among others, also contribute to a *graphics style.* Although this chapter concentrates on the manual formatting of objects, it also shows how to create, apply, modify, and delete graphics styles. If your presentation requires a consistent appearance for its graphic objects, applying graphics styles can save time and prevent errors.

Formatting Lines

In OpenOffice, the term *line* refers to both a freestanding segment and the outer edge of a shape. In most cases, the properties of the line you can modify are its style (solid, dashed, invisible, and so on), its width, and its color. All these options can be applied with a few clicks of the mouse. Select the line you need to format and then use the controls on the Properties deck of the Sidebar, like that shown in Figure 1, or the Line and Filling toolbar to select your desired options.

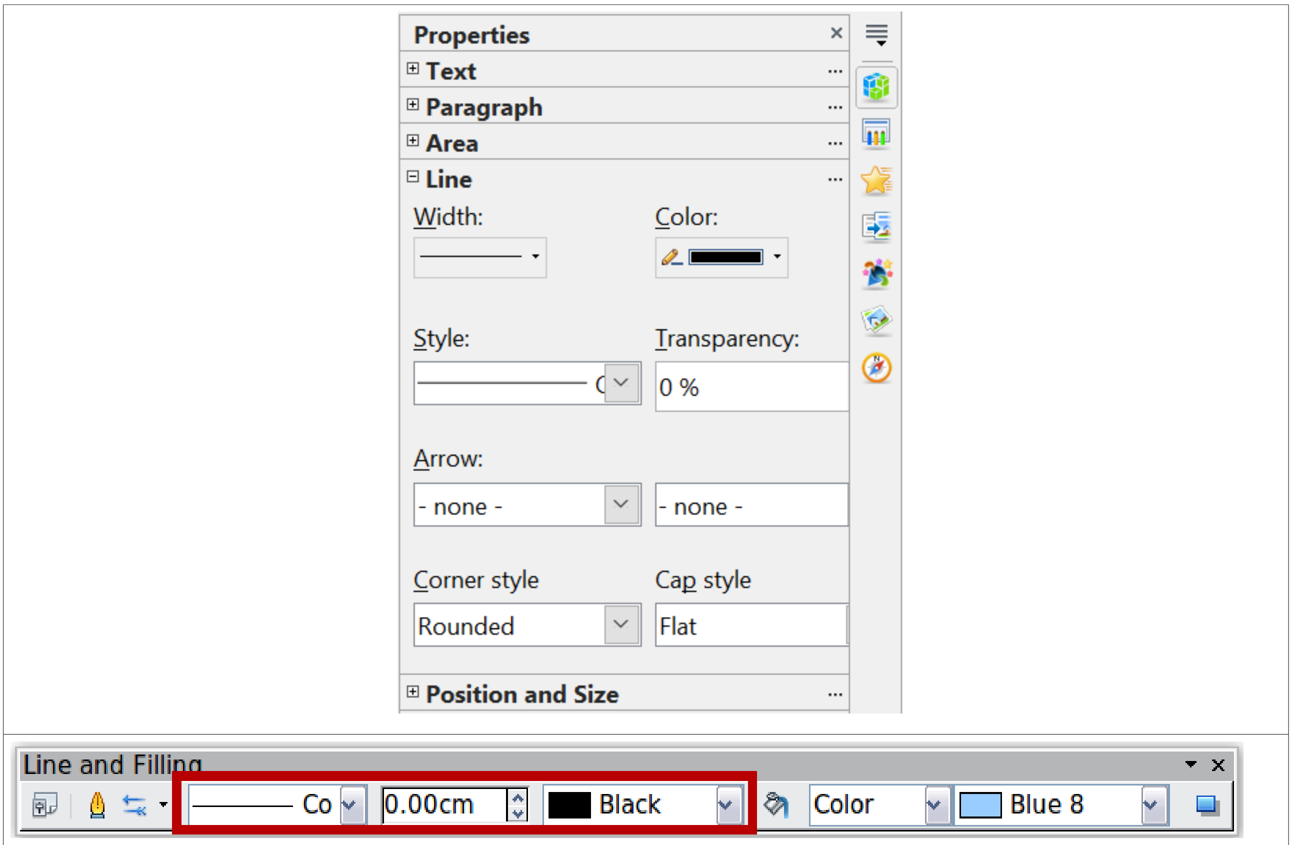


Figure 1: (Top) The Line pane of the Sidebar's Properties deck
(Bottom) Line and Filling toolbar with common line options highlighted

If you need to fine-tune the appearance of a line, choose **Format > Line** from the menu bar, or the More Options button at the top right of the Line pane of the Sidebar, or right-click on the line and select **Line** from the pop-up menu, or select the **Line** icon from the Line and Filling toolbar. These methods open the Line dialog shown in Figure 2, where you can set all the properties of the line at once.

The dialog consists of four tabs: *Line*, *Shadow, Line Styles*, and *Arrow Styles*.

General Line Settings

From the *Line* tab, you can set the basic parameters of the line. This tab is subdivided into four parts:

The *Line Properties* section (on the left side) is the most important. It includes the following parameters:

- **Line style**: A variety of line styles is available in the drop-down list, but more can be defined if needed.
- **Color**: Choose among the predefined colors or refer to "Adding Custom Colors" on page 14 to create a new one.
- **Width**: specifies the thickness of the line.

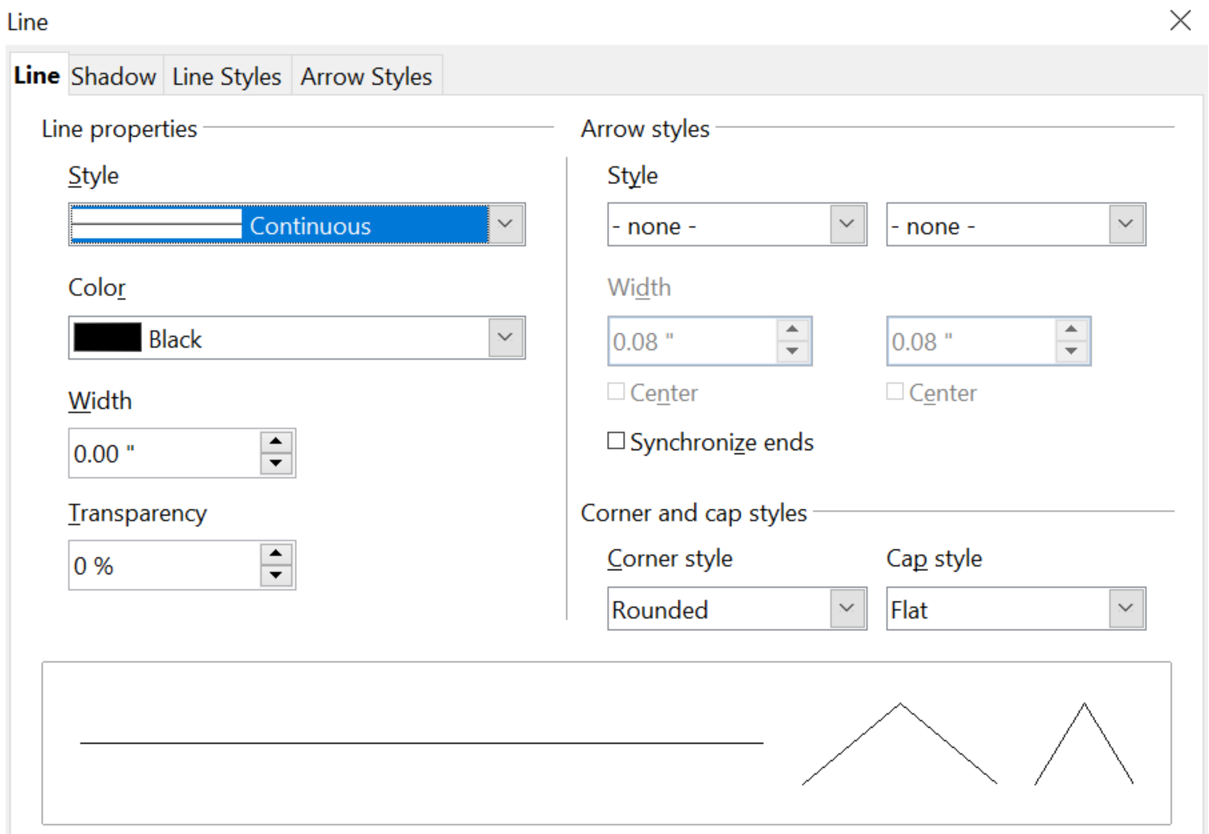


Figure 2: Main line formatting dialog

- **Transparency**: sets the transparency value of the line, a useful property when you don't want to completely hide the background. Figure 3 illustrates the effects on a line of different degrees of transparency.

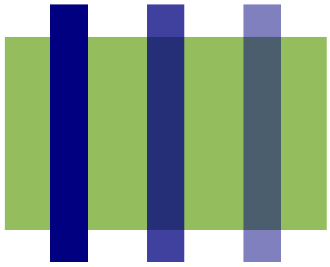


Figure 3: The vertical lines have different levels of transparency (0%, 25%, and 50%).

The *Arrow styles* section of this tab is only applicable to line segments. It has no effect on the line that forms the border of a shape or of a polygon. Use this section to set the styles of the two ends of the segment. You can configure the two ends independently, selecting for each of them the arrow shape (**Style** drop-down menu), the **Width**, and the termination style (**Center** option). Selecting the **Center** option moves the center of the arrowheads to the end point of the line. Figure 4 shows the effects of selecting this option. To make the two ends identical, select the **Synchronize ends** option. To create new arrowheads, use the *Arrow styles* tab, as described in the following section.

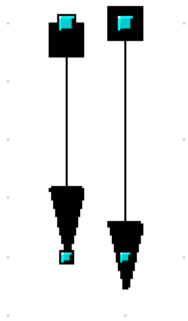
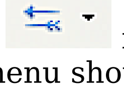


Figure 4: Default arrowheads (left) vs centered arrowheads (right)

The *Corner style* and *Cap style* sections of this tab determine how the connection between two segments and the line ends should look. There are four available options in the *Corner style* drop-down menu and three options in the *Cap style* drop-down. To appreciate the difference between styles, choose a thick line style and observe how the preview changes.

The bottom part of the tab previews the applied style for a single line and two different corners so that the corner and cap style choices can be quickly evaluated.

A faster way to set the arrowheads for a selected line is to click on the **Arrow** drop-down in the Sidebar or the **Arrow Style**  icon in the Line and Filling toolbar (Figure 1). This opens the Arrowheads menu shown in Figure 5, where you can choose one of the many predefined arrowhead styles for the start and termination of the selected segment.

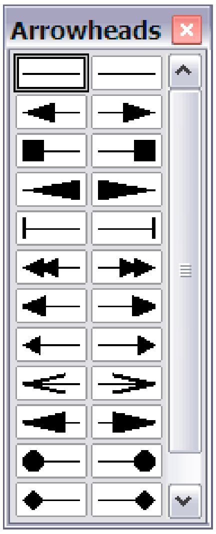


Figure 5: Arrowheads menu

Creating Line Styles

Use the *Line Styles* tab, like that in Figure 6, to create new line styles and load previously saved line styles. Normally, it is not a good practice to modify the predefined styles. Instead, create new ones when necessary.

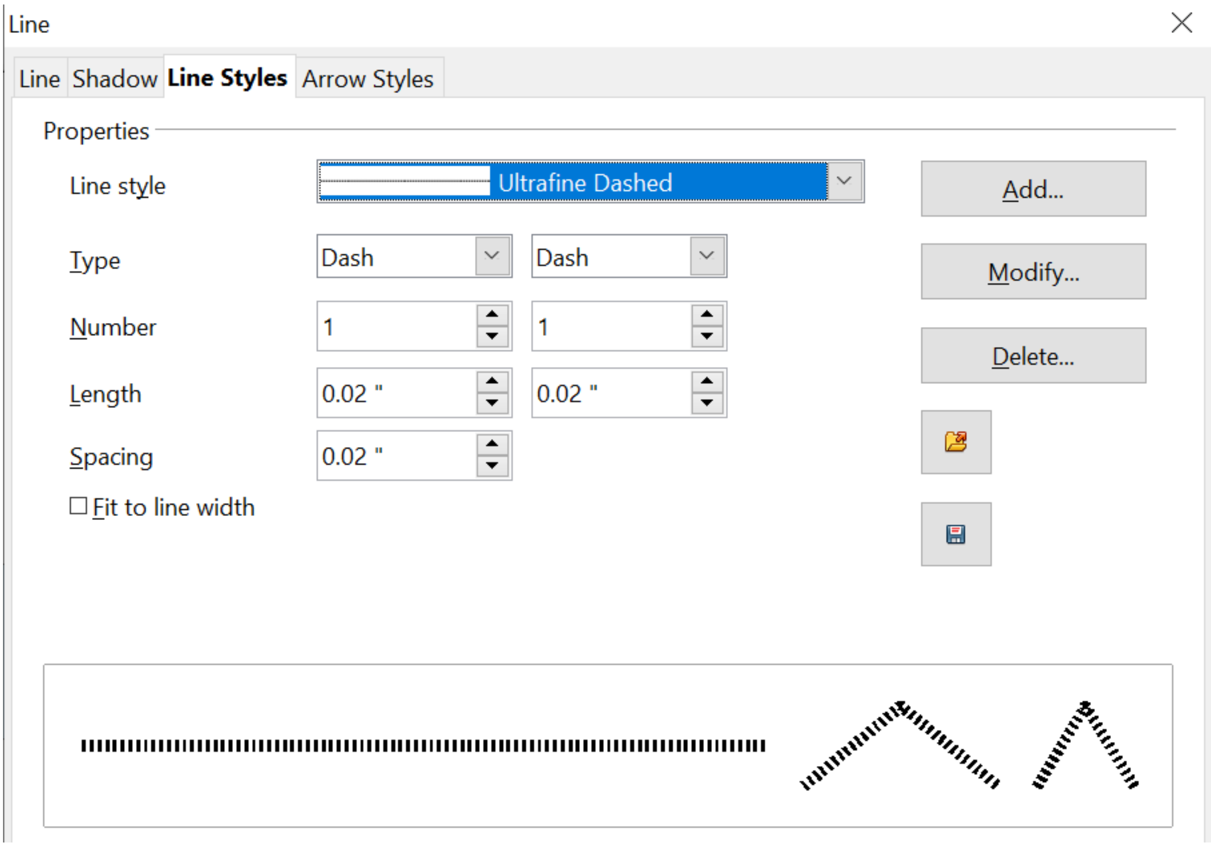


Figure 6: Advanced options for creating line styles

To create a new line style:

1) Choose **Format > Line** from the menu bar.
2) Click on the Line Styles tab.

3) Select from the *Line style* drop-down menu a style similar to the desired one.
4) Click **Add**. On the pop-up dialog, type a name for the new line style and click **OK**.
5) Now define the new style. Start by selecting the line type for the new style. To alternate two line types (for example, dashes and dots) within a single line, select different types in the two **Type** boxes.
6) Specify the number and length (not available for dot style) of each of the types of line selected, set the spacing between the various elements, and decide if the style should fit to the line width (length).

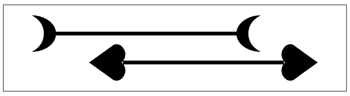
The new line style is available only in the current document. If you want to reuse the line style in other presentations, click the **Save Line Styles** icon and type a descriptive name. This saves all of the line styles in this presentation. (Saved styles have a file extension of .sod.)

To make previously saved line styles available in the current presentation, click the **Load Line Styles** icon, select the saved list of styles, and click **Open**.

Use the **Modify** button to change the name of the style.

Creating Arrow Styles

Use the fourth tab of the Line dialog to create new arrow styles, such as the ones in the figure below, to modify existing arrow styles, or load previously saved arrow styles.



1) First, draw a curve with the shape you want for the arrowhead.

Note	The arrowhead must be a *curve*. A curve is something you could draw without lifting a pencil. For example, is a curve but is not a curve. You can however draw shapes which are not curves and then at the end convert them to a curve.

The top part of the shape will point in the direction of the line. In Figure 7 the corner at the top of the shape will point towards the "outside" of the line.

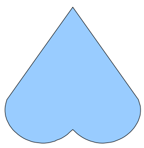


Figure 7: To create your own arrowhead, first draw a curve.

2) Select the curve. With the resizing handles showing, select **Format > Line** from the menu bar, or right-click and choose **Line** from the pop-up menu.
3) Go to the *Arrow styles* tab (Figure 8), click the **Add** button, type a name for the new arrow style, and click **OK**.
4) You can now access the new style from the Arrow style list. When you select the name of the new style, it is shown at the bottom of the dialog.

Line Shadow

Use the *Shadow* tab of the Line dialog to add and format the line shadow. The settings on this tab are the same as those for shadows applied to other objects and are described in “Formatting Shadows” on page 21.

A faster way to apply a shadow to the line is using the last button of the Line and Filling toolbar of Figure 1. The main disadvantage of using the toolbar button is that the shadow appearance will be constrained by the shadow settings of the default graphics style.

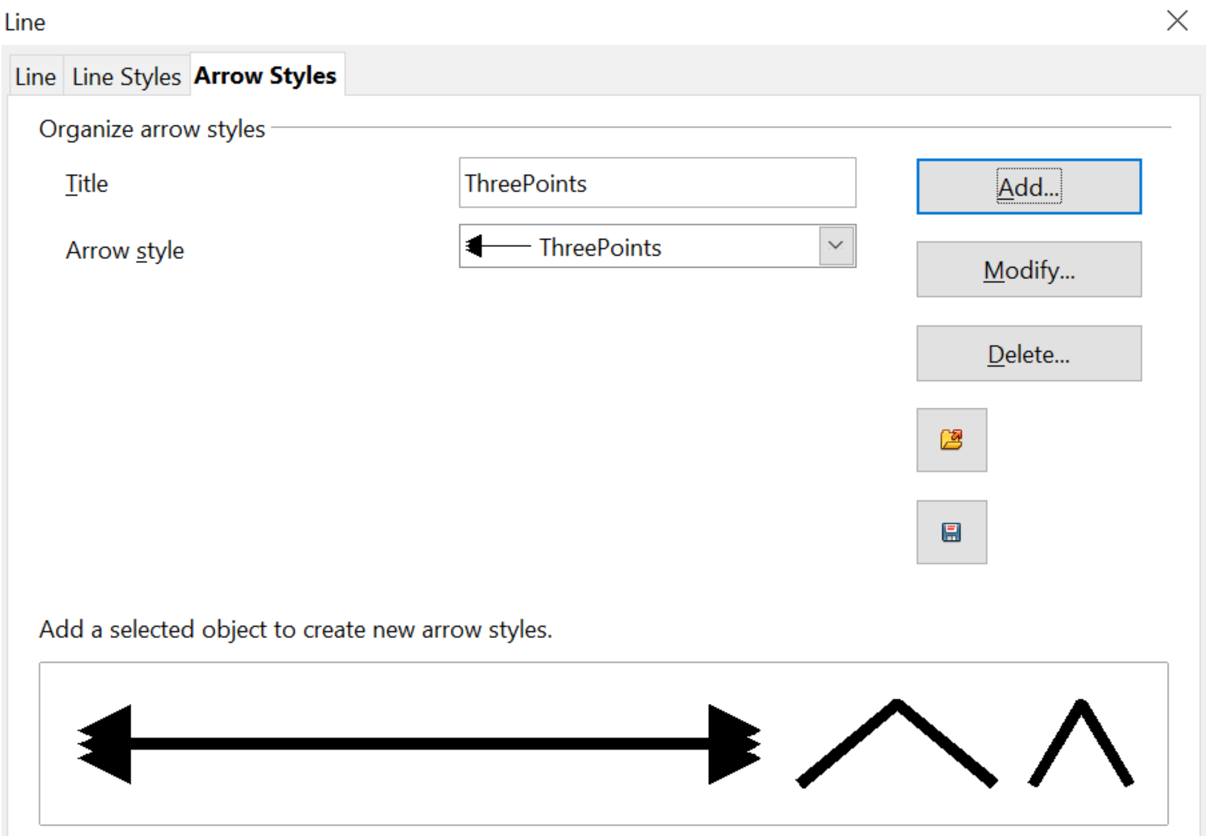


Figure 8: Advanced options for creating arrow styles

Formatting the Fill Area

The term **area fill** refers to the inside of an object, which can be a uniform color, a gradient, a hatching pattern, or an image. An area fill, such as that shown in Figure 9, can be made partly or wholly transparent and can throw a shadow.

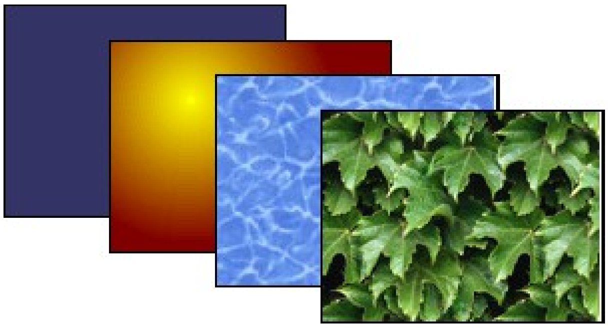


Figure 9: Different types of area fill

The Line and Filling toolbar (Figure 10) has the majority of the tools normally used to format graphic objects. If this toolbar is not showing, choose **View > Toolbars > Line and Filling** from the menu bar. You can also use the Area dialog, described on page 12.

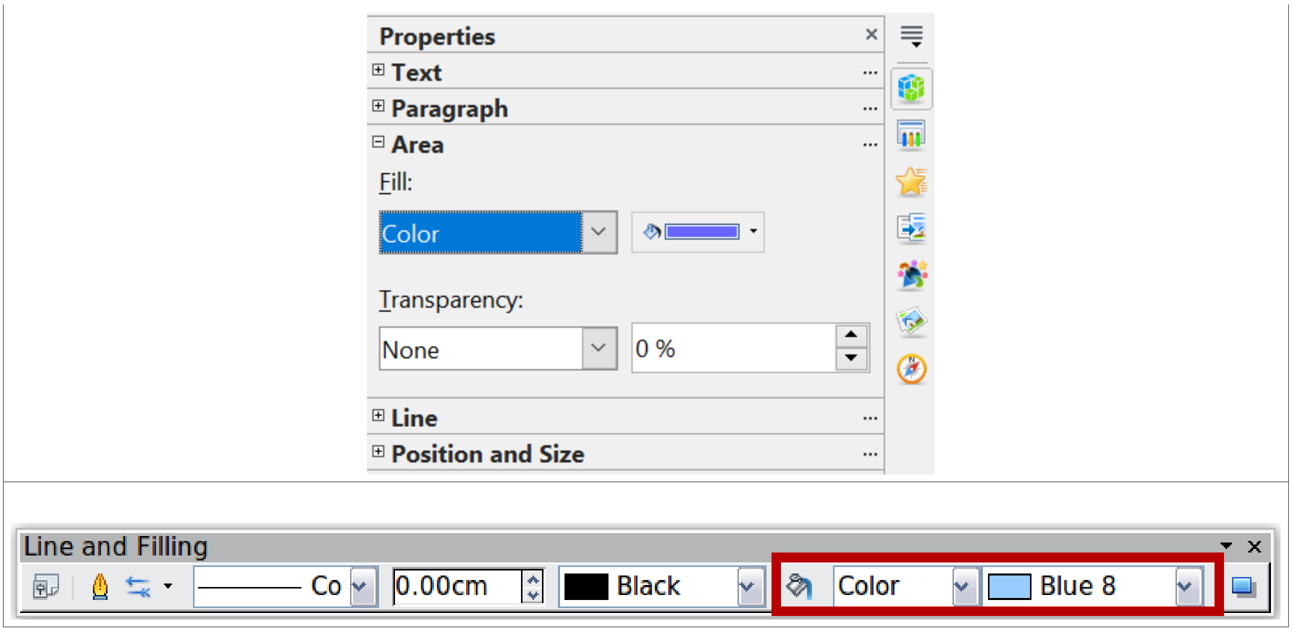
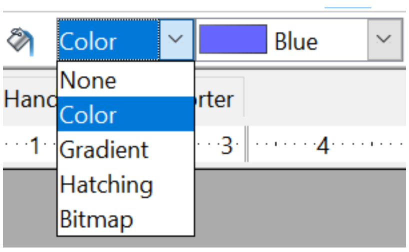


Figure 10: Common fill options on the Area pane of the Sidebar (top) and the Line and Filling toolbar (bottom)

To format the area of an object, select it so that the green resizing handles show. A wide number of default fillings are readily available from the Line and Filling toolbar. First, select the type of fill from the pull-down menu to the right of the paint can icon. If you want no fill at all, select *None*.



Once you have decided on a predefined or custom fill, you can further refine it by adding a shadow or transparency.

Uniform Color

Select the object you wish to edit. On the Sidebar or the Line and Filling toolbar, select **Color** from the pull-down list, and then choose a color from the drop-down menu, like that in Figure 11.

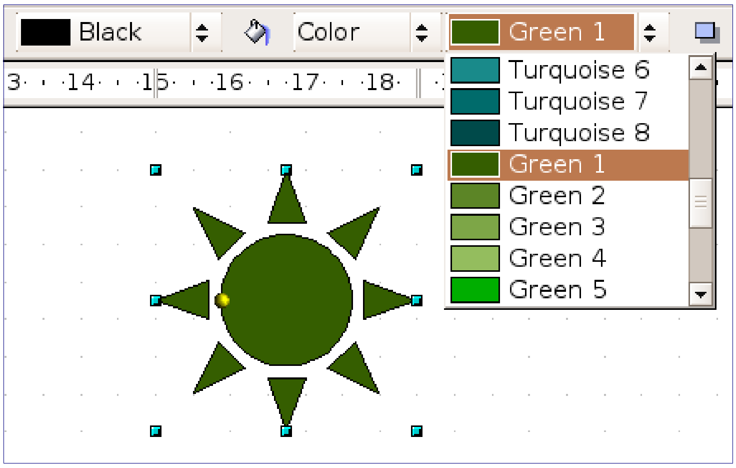


Figure 11: Filling with a color

Fill with a Gradient

A gradient fill provides a smooth transition from one color to another. The transition pattern may vary from a simple linear transition to a more complex radial transition.

Select the object you wish to edit. On the Sidebar or Line and Filling toolbar, select **Gradient** and then choose a gradient from the drop-down menu, like that in Figure 12.

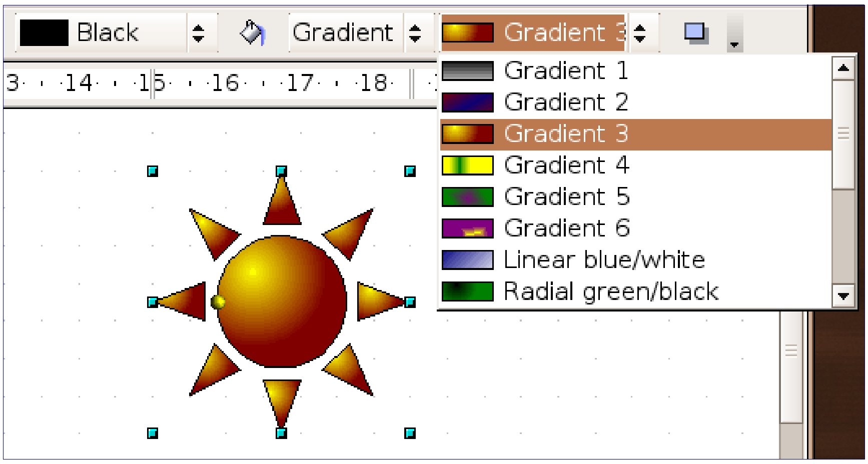


Figure 12: Filling with a gradient

Fill with a Line Pattern (Hatching)

Select the object you wish to edit. On the Sidebar or Line and Filling toolbar, select **Hatching**. Then choose a hatching fill, like that shown in Figure 13, from the drop-down menu. A hatching fill is applied throughout the area.

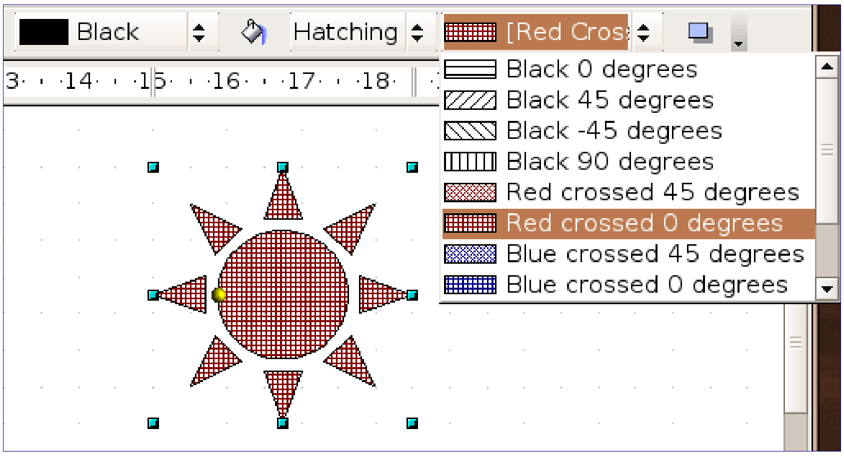


Figure 13: Filling with a line pattern

Fill with an Image

You can fill an object only with a bitmap image, that is, a binary digit map that functions as a digital image file format. Bitmaps represent a graphic using pixels, each pixel having its own color and collectively forming the image. No vector graphic image, such as an image created using mathematical formulas to represent the image, can be used here. Select the object you wish to edit. On the Sidebar or Line and Filling toolbar, select **Bitmap**, as shown in Figure 14, and then choose a bitmap fill from the drop-down menu.

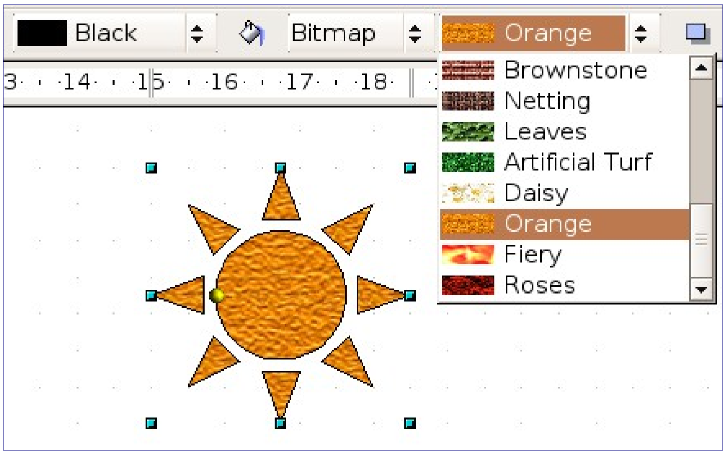


Figure 14: Filling with an image

Using the Area Dialog

In addition to using the Line and Filling toolbar, you can use the Area dialog to apply existing fills and create your own. To open it, choose **Format > Area** from the menu bar, or click the three dots at the top right corner of the Area pane of the Sidebar, or click on the paint bucket icon on the Line and Filling toolbar, or right-click on the object and select **Area**, like that in Figure 15.

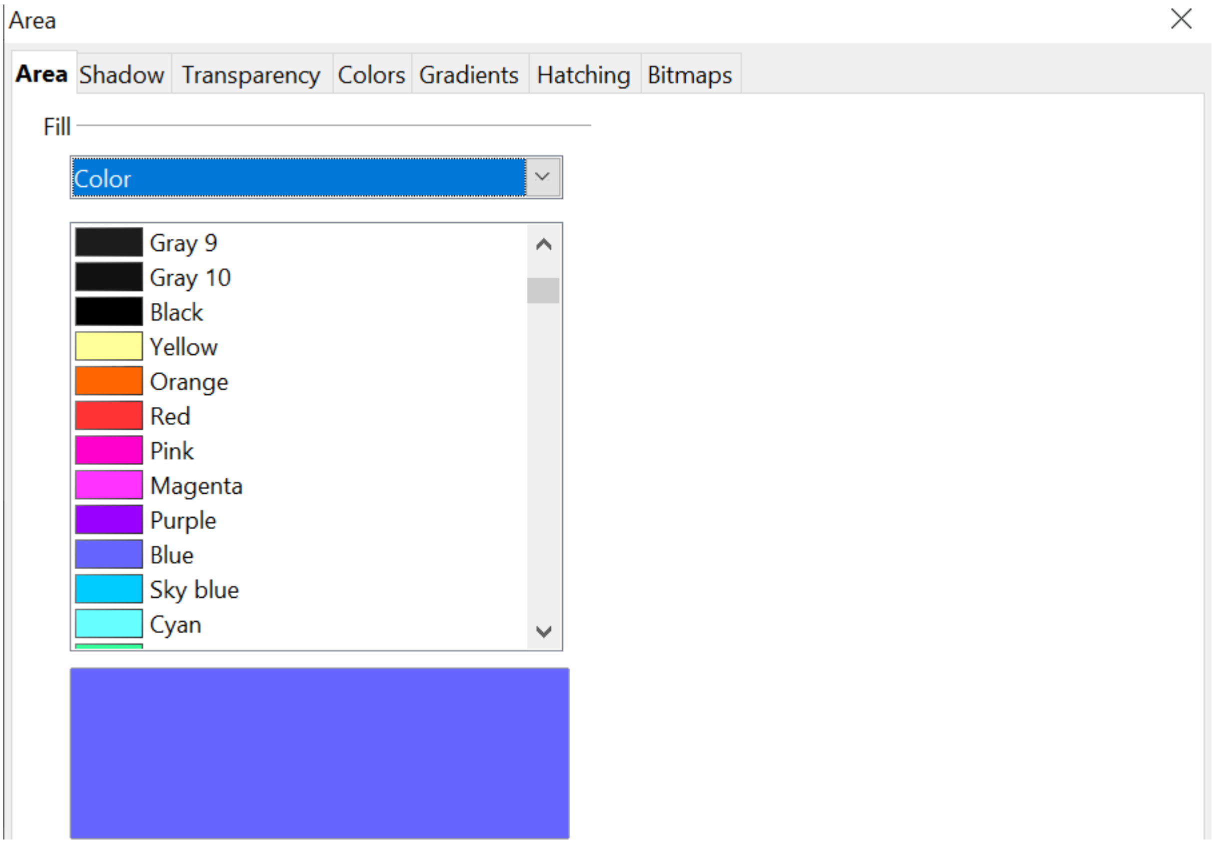


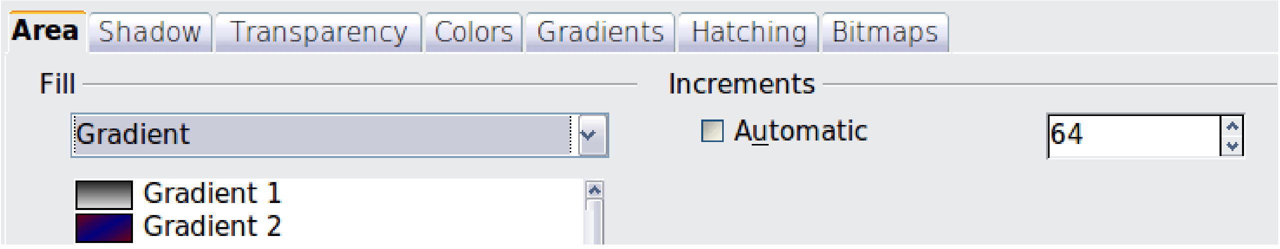
Figure 15: Area tab of the area formatting dialog

Use the *Area* tab to apply predefined fills, both those supplied with OpenOffice and those you create yourself. Use the *Colors*, *Gradients*, *Hatching*, and *Bitmaps* tabs to define new fills, as described in "Creating New Area Fills" on page 14. The *Transparency* tab is discussed on page 22. To make the object cast a shadow, see page 21.

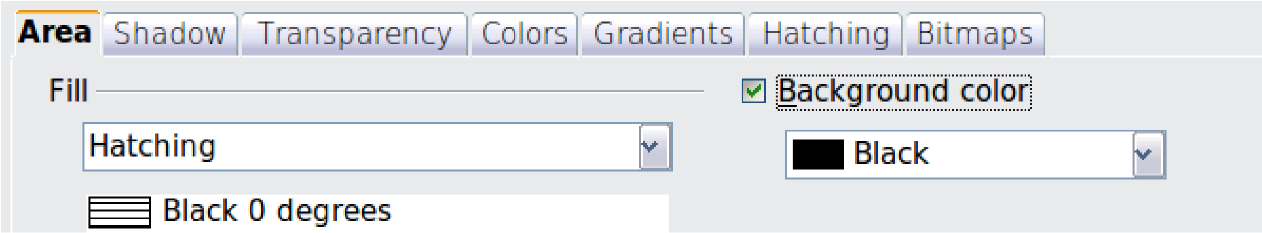
To apply an area fill, first select the required fill type from the top-left drop-down list. The tab changes to show the list of predefined options for that fill type in the middle section.

When using the Area tab of the Area dialog, additional options may become available after selecting the fill type and one of the available fill options.

- For gradient fills, you can override the number of steps (increments) that should be applied to the transition from one color to the other. To do so, select Gradient on the Area tab and deselect the **Automatic** option under *Increments.* Then enter the number of steps required in the box to the right.



- For hatching, you can apply a different background color by selecting the **Background color** option and choosing a color from the drop-down list.



- For bitmaps, you can customize a large number of parameters. Refer to “Working with Bitmap Fills” on page 18 for additional information.

Creating New Area Fills

The following sections describe how to create new fills and how to apply them.

Although you can change the characteristics of an existing fill and then click the **Modify** button, it is recommended that you create new fills or modify custom fills rather than the predefined ones. Pre-defined fills may be reset when updating OpenOffice.

Adding Custom Colors

On the *Colors* tab, shown in Figure 16, you can modify existing colors or create your own.

You can specify a new color either as a combination of the three primary colors, Red, Green, and Blue (RGB notation) or by the percentage of Cyan, Magenta, Yellow, and Black (CMYK notation).

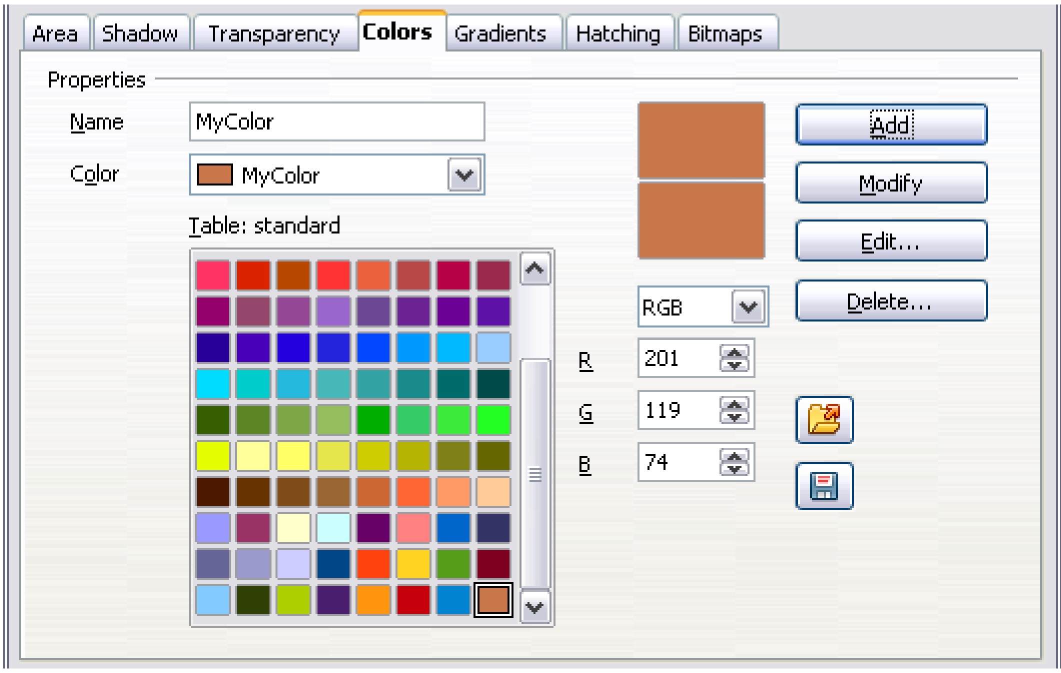


Figure 16: A custom color has been added to the list

To create a new color:

1) Enter the name for the color in the *Name* box.

2) Select whether to define the color in RGB or CMYK. For RGB, specify the Red, Green, and Blue components on a 0 to 255 scale. For CMYK, specify the Cyan, Magenta, Yellow, and Black (K) components, from 0% to 100%.
3) Click the **Add** button. The color is now added to the list on the Area tab.

To modify a color:

1) Select the color to modify from the list.
2) Enter the new values that define the color (if necessary, change the settings between RGB and CMYK).
3) Modify the name as required.
4) Click the **Modify** button.

Alternatively, use the **Edit** button (this will open a new dialog), modify the color components as required, and click **OK** to exit the dialog.

Use the Load and Save buttons in the dialog to use a different color palette or to save your own custom colors.

Tip	You can also add custom colors using **Tools > Options > OpenOffice > Colors**. This method makes the color available to all components of OpenOffice, whereas colors created using the Area dialog are only available for Impress.

Creating Gradients

To create a new gradient or to modify an existing one, select the Gradients tab shown in Figure 17 from the Area dialog. Several types of gradients are predefined, and in most cases, changing the *From* and *To* colors will be sufficient to obtain the desired result.

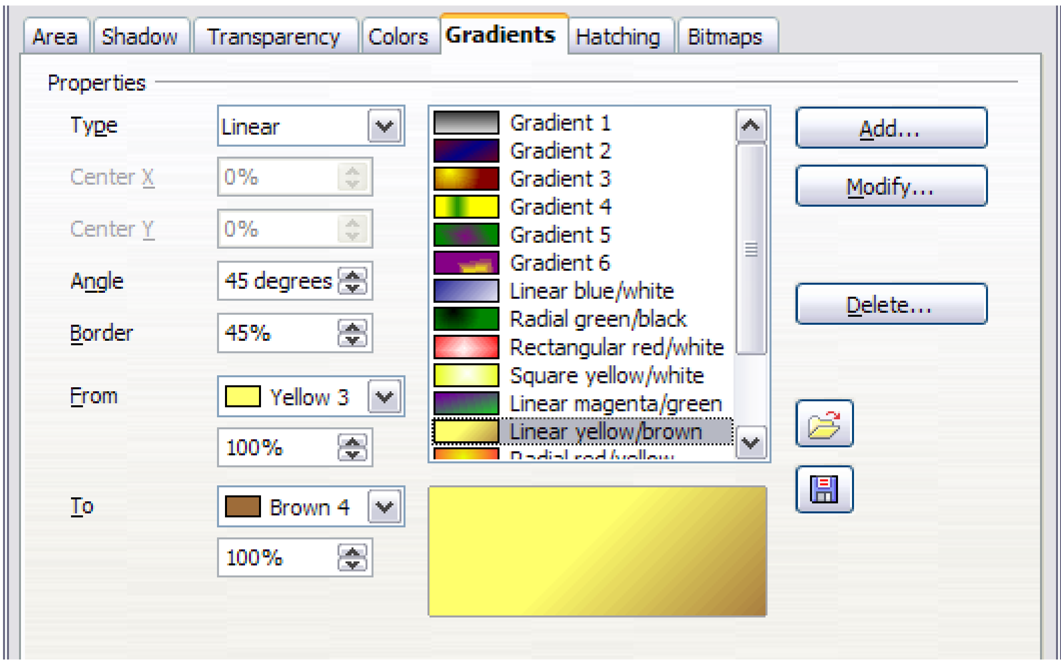


Figure 17: The Gradients tab of the Area dialog

It is highly recommended that you create a new gradient even if you just want to change two colors. Predefined colors should be used only as starting points.

To create a new gradient:

1) First, choose the *From* and *To* colors with the tool shown in Figure 18.

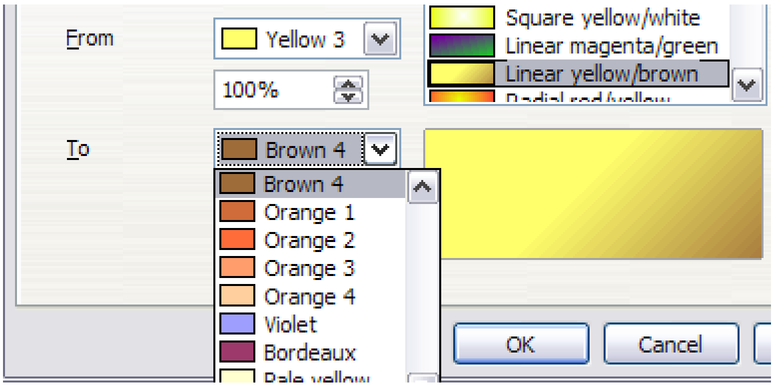


Figure 18: Gradient transition color selection

2) Then choose a type of gradient from the list: Linear, Axial, Radial, Ellipsoid, Square, or Rectangular, as is illustrated in Figure 19.

   A preview of the gradient type is shown under the available gradients list in the middle of the dialog. Figure 17 shows an example.

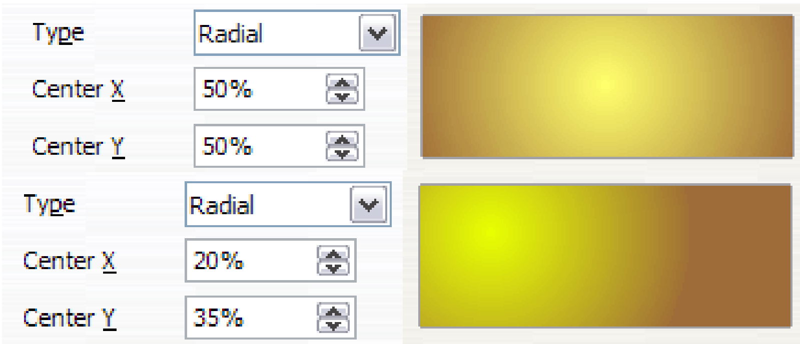


Figure 19: Center option in a radial gradient

3) Depending on the chosen type, some options will be grayed out. Set all the properties as desired (very often, the default values will work well). The properties to set to create a gradient are summarized in Table 1.
4) Click the **Add** button to add the newly created gradient to the list.

Tip	The newly created gradients remain available to all OpenOffice components and also for future presentations. It pays to give them descriptive names.

Table 1: Gradient properties

Property	***Meaning***
Center X	For Radial, Ellipsoid, Square, and Rectangular gradients, modify these values to set the horizontal offset of the gradient center.
Center Y	For Radial, Ellipsoid, Square, and Rectangular gradients, modify these values to set the vertical offset of the gradient center.
Angle	For all the gradient types, specifies the angle of the gradient axis.

Property	Meaning
Border	Increase this value to make the gradient start further away from the border of the shape.
From	The start color for the gradient. In the edit box below, enter the intensity of the color: 0% corresponds to black, 100% to the full color.
To	The end color for the gradient. In the edit box below, enter the intensity of the color: 0% corresponds to black, 100% to the full color.

Creating Hatching Patterns

You can create new hatching patterns or modify existing ones. Start by selecting the *Hatching* tab of the Area dialog, shown in Figure 20.

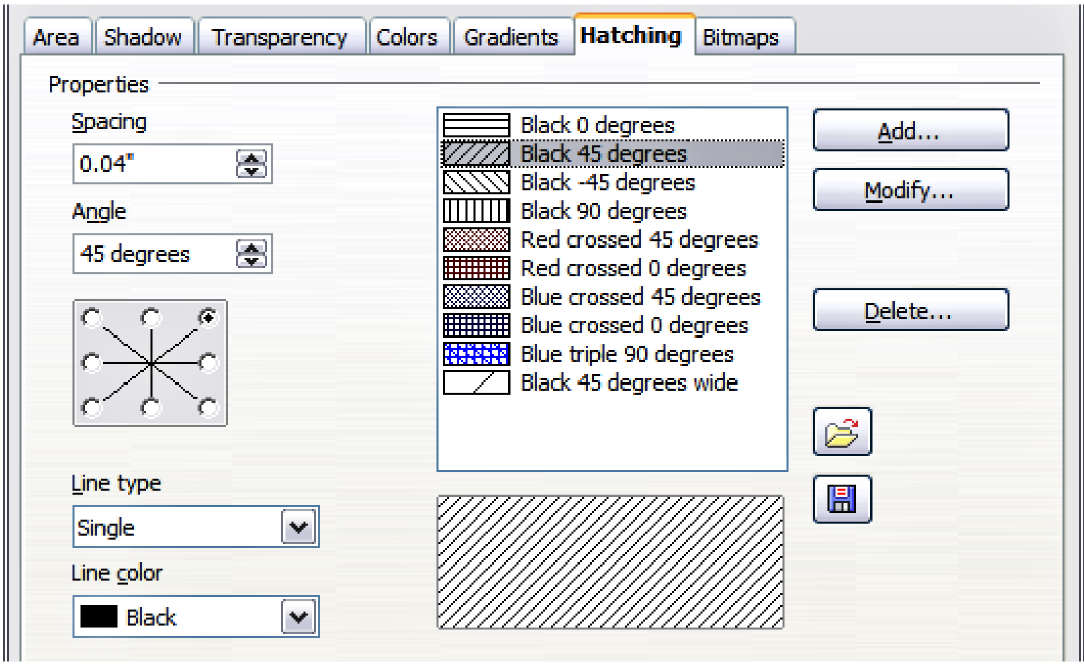


Figure 20: The Hatching tab of the Area fill dialog

As with gradients and colors, it is better to create a new pattern than to modify a predefined one.

To do so:

1) Select a pattern similar to the one that will be created as a starting point.
2) Modify the properties of the lines forming the pattern. A preview is displayed in the window below the available patterns.
3) Click the **Add** button and choose a name for the newly created hatching.

The properties that can be set for a hatching pattern are shown in Table 2.

Table 2: Properties of hatching patterns

Property	**Meaning**
Spacing	Determines the spacing between two lines of the pattern. As the value is changed, the preview window is updated.
Angle	Use the mini map below the numerical value to quickly set the angle formed by the line to multiples of 45 degrees. If the required angle is not a multiple of 45 degrees, enter the desired value in the edit box.
Line type	Set single, double, or triple line for the style of the pattern.
Line color	Use the list to select the color of the lines that will form the pattern.

Working with Bitmap Fills

On the Area tab, choose *Bitmap* from the drop-down list. Select the bitmap to be used to fill the area from the list. Note that any imported bitmaps will become available in the list.

Set the size, position, and offset parameters (as applicable) on the right side of the tab, and then click **OK** to close the dialog.

As Figure 21 shows, there are several parameters to be configured when using a bitmap fill. These are described in Table 3.

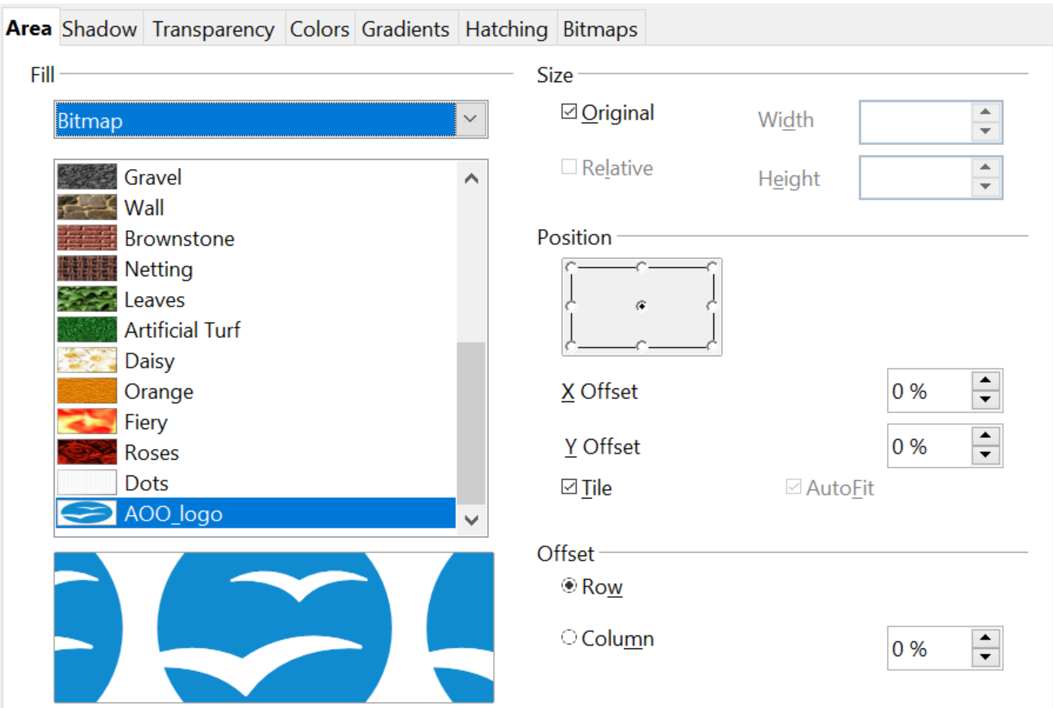


Figure 21: Advanced formatting for bitmap fill

Table 3: Bitmap fill properties

Property	**Meaning**
Size – Original	Select this box to retain the original size of the bitmap.

Property	***Meaning***
Size – Relative	To rescale the object, deselect the Original option and select this one. The Width and Height edit boxes are enabled.
Size – Width	When Relative is selected, 100% means that the bitmap's original width will be resized to occupy the whole fill area width, 50% means that the bitmap's width will be half that of the fill area.
Size – Height	When Relative is selected, 100% means that the bitmap's original height will be resized to occupy the whole fill area height, 50% means that the bitmap's height will be half that of the fill area.
Position – Anchor Map	Select from the map the place within the area to which the bitmap should be anchored.
Position – Tile	When this option is selected, the bitmap will be tiled to fill the area. The size of the bitmap used for the tiling is determined by the Size settings.
Position – X offset	When Tile is enabled, enter in this box the offset for the width of the bitmap in percentage values. 50% offset means that Impress will place the middle part of the bitmap at the anchor point and start tiling from there.
Position – Y offset	This will have a similar effect to the X offset, but will work on the height of the bitmap.
Position – Autofit	Stretches the bitmap to fill the whole area. Selecting this option disables all the size settings.
Offset – Row	If Tile is enabled, it offsets the rows of tiled bitmaps by the percentage entered in the box so that two subsequent rows are not aligned.
Offset – Column	If Tile is enabled, it offsets the columns of tiled bitmaps by the percentage entered in the box so that two subsequent columns of bitmaps are not aligned.

The best way to acquire an understanding of these parameters is to use them. Figure 22 shows some examples of bitmap fills and the parameters used.

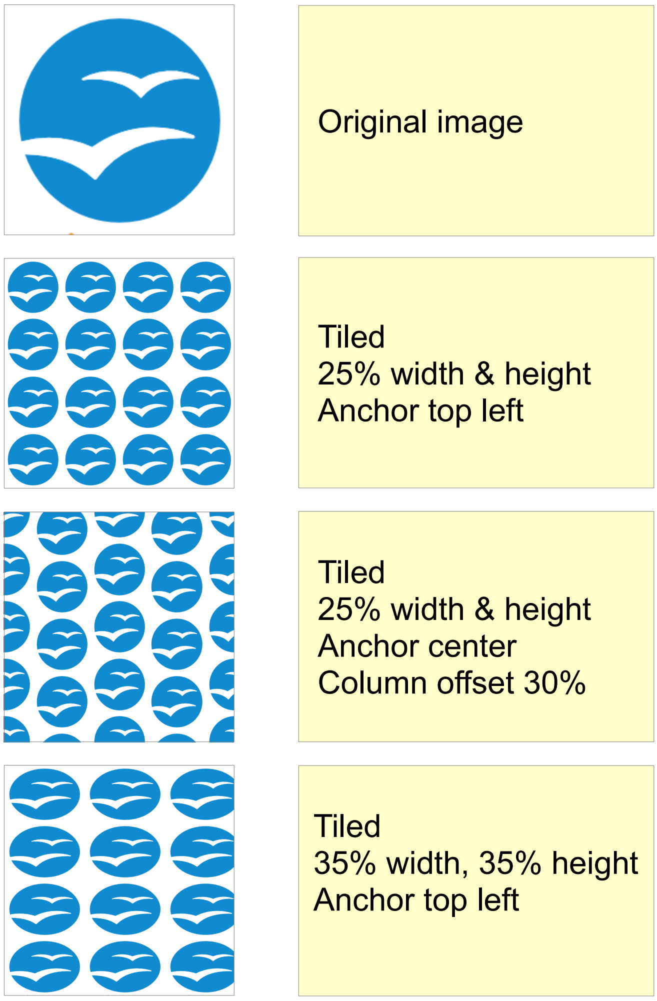


Figure 22: Examples of bitmap fill

Creating and Importing Bitmaps

You can add (import) new bitmap fills or create your own pattern on an 8x8 grid, using the *Bitmaps* tab of the Area dialog.

To create a bitmap fill:

1) Start with the Blank bitmap type at the top of the list to activate the Pattern editor, as shown in Figure 23.
2) Select the Foreground and Background colors. The square in the Pattern editor will fill with the chosen color.
3) Start creating the pattern by clicking the squares (pixels) with the left mouse button that you want to be painted in the background color. Check the preview window to see if the desired effect is achieved.
4) When done, click **Add** to save the pattern.

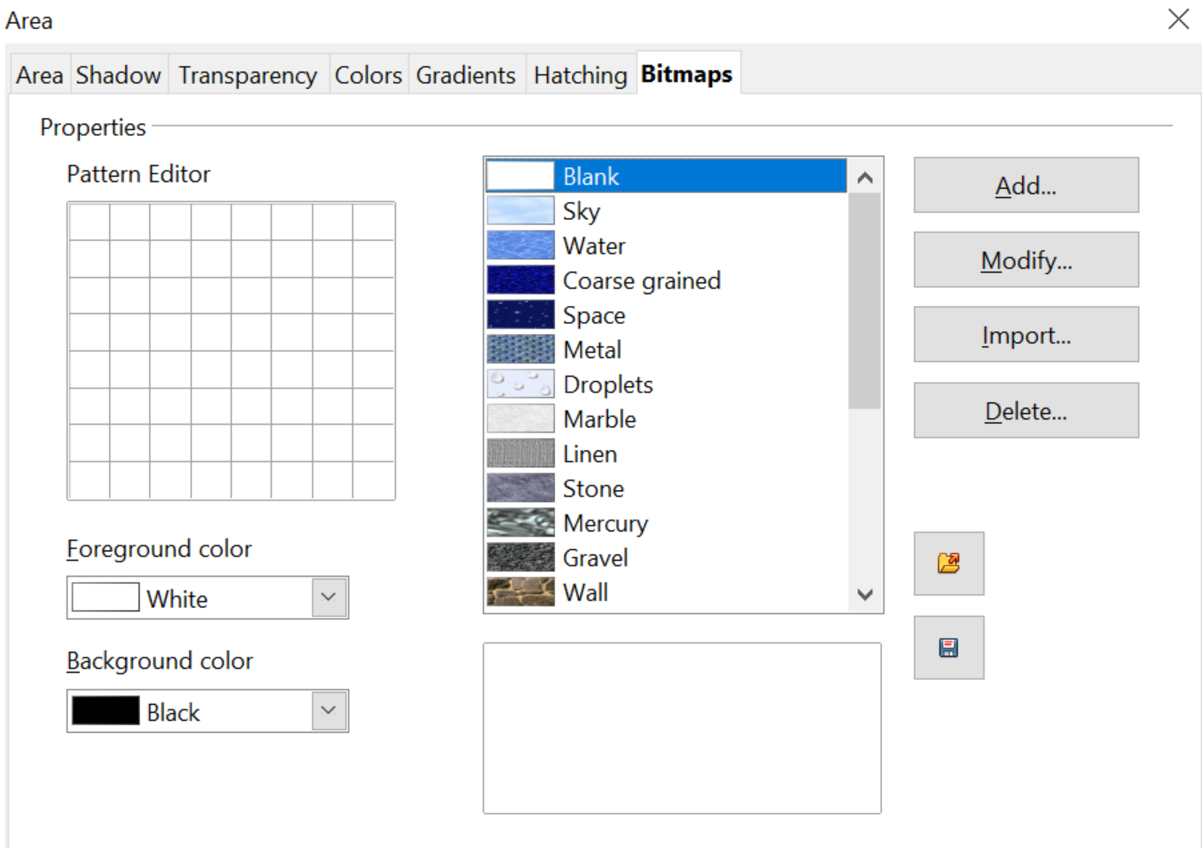


Figure 23: The Bitmaps tab of the Area fill dialog.

To import a bitmap created in Draw or another program:

1) Click the **Import** button.
2) A file picker dialog is displayed. Browse to the directory containing the bitmap file, select it, and then click **Open**.
3) Type a name for the imported bitmap and click **OK**.

Note	Bitmaps generally have an extension .bmp or .png. To create a bitmap image with Draw, select **File > Export**, choose **PNG** from the pull-down list of file formats, give the file a name, and save it.

Formatting Shadows

Shadowing can be applied to both lines and areas. To apply a shadow to an area, first select the object to which shadowing should be applied, then select **Format > Area.**

Shadows can also be applied to lines. One way is to click the **Shadow** icon  on the Line and Filling toolbar (the last tool on the right-hand end). Alternatively, you can apply a style to the line that uses a shadow (see “Working with Graphics Styles” on page 28 for additional information on using styles).

Using the first method, you cannot customize the shadow; it is applied according to the default settings. The second method offers the opportunity to configure the shadow properties.

The dialog to customize a shadow is shown in Figure 24.

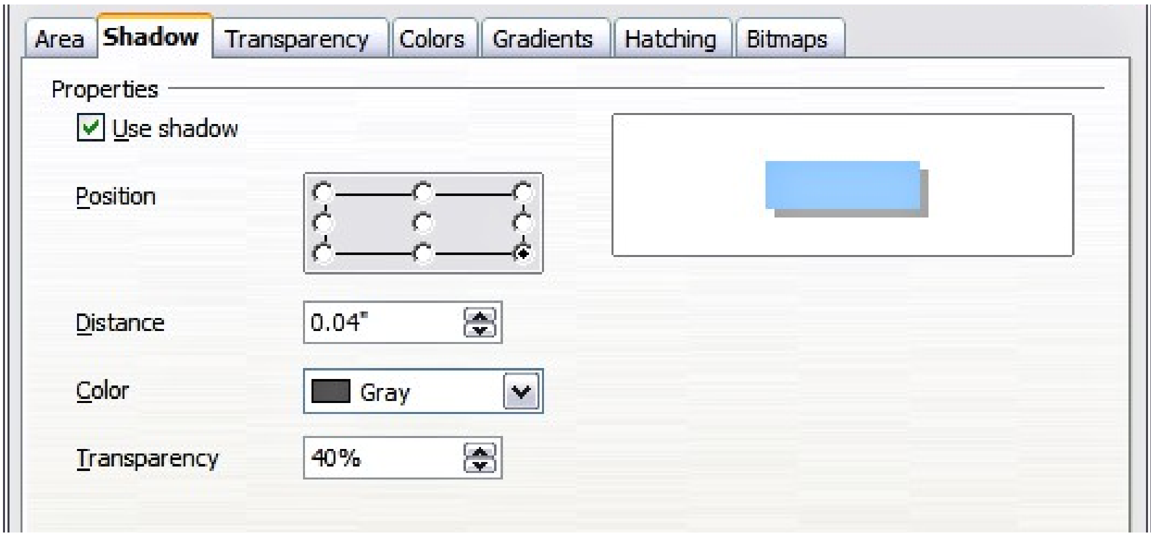


Figure 24: Dialog for customizing the shadowing of graphic objects

When the **Use shadow** option is selected, the following parameters can be set:

- **Position**: The selected point in the mini-map determines the direction in which the shadow is cast.
- **Distance**: determines the distance between the object and the shadow.
- **Color**: sets the color of the shadow.
- **Transparency**: determines the amount of transparency for the shadow.

Tip	When the transparency value is set above 0%, the shadow does not completely hide the objects below. This produces a pleasant visual effect, as shown in Figure 25.

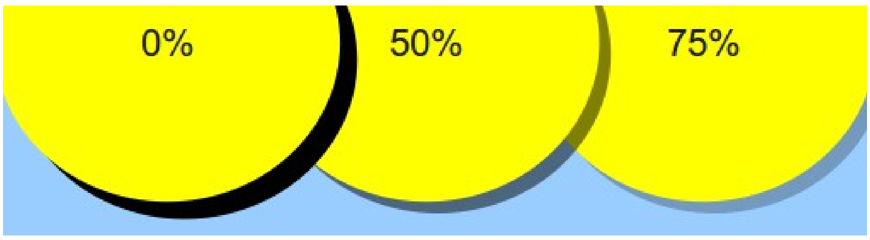


Figure 25: Shadows with different levels of transparency

Transparency Formatting

Transparency is also applicable to lines, areas, and shadows. To apply transparency to lines, refer to “Formatting Lines” on page 4; for shadows, refer to “Formatting Shadows” above.

To apply transparency to areas, select **Format > Area** and then go to the Transparency tab, as shown in Figure 26.

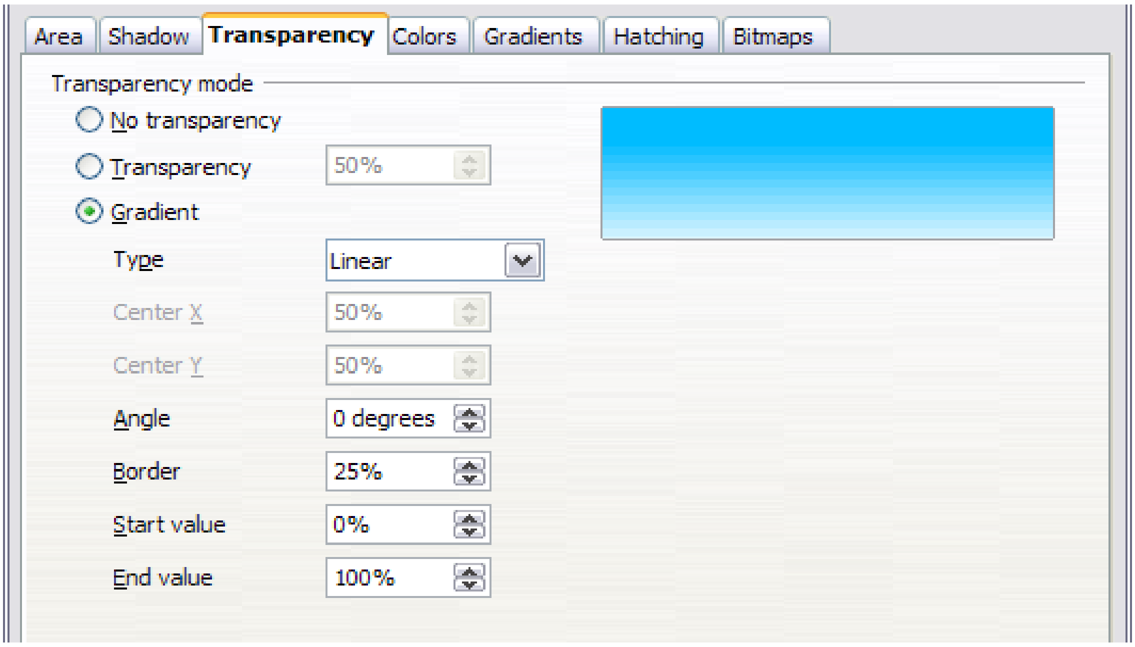


Figure 26: Setting the object transparency

The two types of transparency are *uniform transparency* and *gradient transparency*. To obtain uniform transparency, select **Transparency** and then select the percentage of transparency required. For a gradient transparency (so that the area becomes gradually transparent), select **Gradient** and then set the parameters of the gradient. Refer to Table 1 on page 16 for a description of most of the settings. In the Transparency tab, specify the Start value and the End value for the transparency gradient. 0% is fully opaque, 100% means fully transparent.

An example of gradient transparency is shown in Figure 27.

For more information on gradient transparency, including an example of combining a color gradient with gradient transparency, refer to “Advanced Gradient Controls” on page 24.

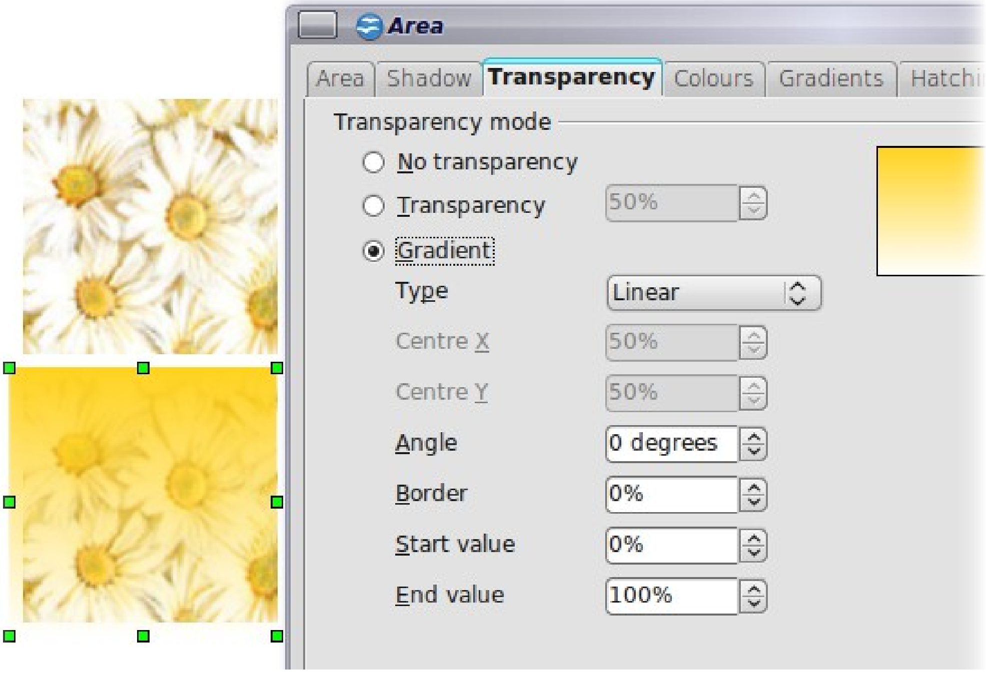
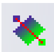


Figure 27: Example of gradient transparency

Advanced Gradient Controls

As discussed in "Creating Gradients" on page 15 and "Transparency Formatting" on page 22, gradient properties can be configured using the parameters in Table 1 on page 16.

Impress also provides a graphical interface for modifying the gradient parameters using only the mouse. To use these tools, click on the **Transparency** icon  or the **Gradient** icon  in the Mode toolbar shown in Figure 28.

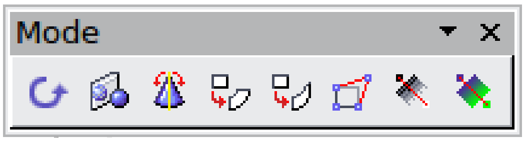


Figure 28: The Mode toolbar

When an object with a gradient fill is selected, click on the Gradient icon to display a dashed line connecting two squares colored as the "From" color and the "To" color of the gradient, as shown in Figure 29 and Figure 30 below.

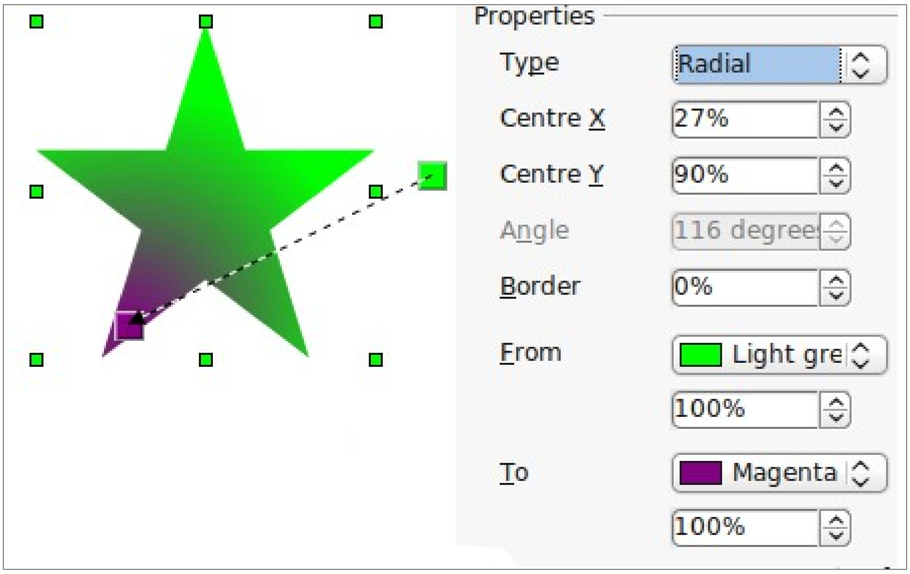


Figure 29: Green to magenta gradient

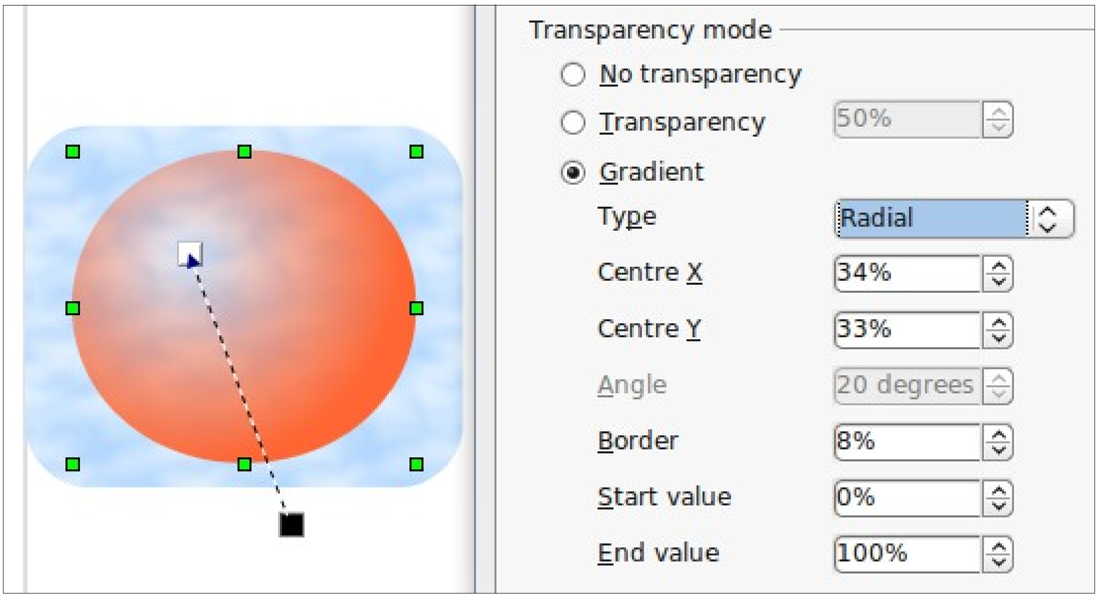


Figure 30: Transparency gradient (note how the background becomes visible close to the white square)

For **linear gradients**, move the square corresponding to the *From* color to change where the gradient starts (that is, the Border property). Move the square corresponding to the *To* color to change the orientation (Angle property).

For **axial gradients**, you can move only the *To* color to change both the angle and the border properties of the gradient.

For **radial gradients**, move the *From* color to modify the border property (that is, how “wide” the gradient circle is). Move the *To* color to change the point where the gradient ends (Center X and Center Y values).

For **ellipsoid gradients**, move the *From* color to modify the border property (the size of the gradient ellipsoid). Move the *To* color to change the angle of the ellipsoid axis and the axis itself.

For **square and rectangular gradients**, move the *From* color to modify the border (the size of the gradient square or rectangle) and the angle of the gradient shape. Move the *To* color to change the center of the gradient.

The same actions can be performed for transparency gradients, with the difference that the two squares activated by the Transparency icon represent the fully opaque point (black square) and the fully transparent point (white square).

These icons are grayed out by default and are only activated when an object with gradient filling, gradient transparency, or both is selected.

In both cases, a dashed line connecting two small squares appears on top of the object. Click outside the object to set the gradient.

Note	Moving the squares will have different effects depending on the type of gradient. For example, for a linear gradient, the start and end squares of the gradient will always be situated to either side of the center point of the object.

Formatting Text

Impress provides two dialog boxes related to text formatting: **Format > Character** and **Format > Text**.

To modify formatting such as font and font effects, select the text in the shape and then go to **Format > Character**. For more information, see Chapter 3 (Adding and Formatting Text). This section covers the formatting of the overall shape of text, which is added to a line or a shape.

To add text to an object (a shape or a line):

1) Select the object to which text will be added.
2) With the green resizing handles showing, double-click on the object and wait for the cursor to become an I-beam, or just start typing.
3) Type the text. When finished, click somewhere outside the object or press *Esc.*

To format the text in a shape:

1) Select the object to which text was added.
2) Select **Format > Text** or right-click on the shape and select **Text** from the pop-up menu. The Text dialog is displayed.

The top section of the *Text* tab, shown in Figure 31, offers several options in the form of checkboxes. Some of the options will be grayed out, depending on the object to which the text will be attached.

- Select **Fit width to text** to expand the width of the shape or line if the text is too long for it.
- Select **Word wrap text in shape** to start a new line automatically when the edge of the shape is reached.
- Select **Fit height to text** to expand the object height whenever it is smaller than the text (set by default for lines).
- Select **Resize shape to fit text** to expand a custom shape when the text inserted in the shape is too large.
- Select **Fit to frame** to expand the text so that it fills all the available space.
- Select **Adjust to contour** to make the text follow a curved line.

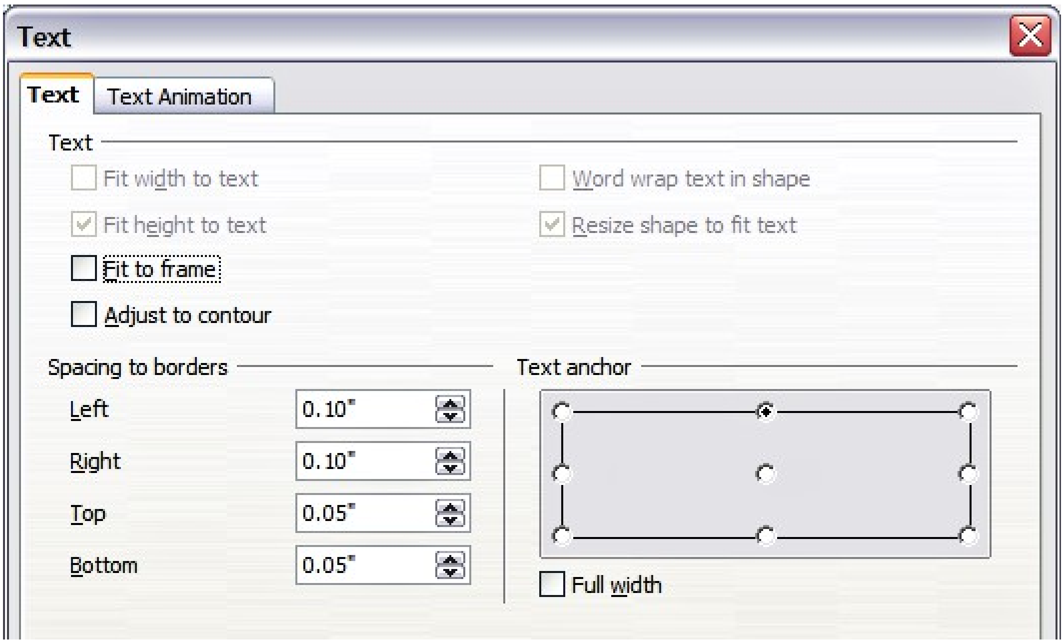


Figure 31: Main dialog to set the text properties

In the *Spacing to borders* section, specify the amount of space to be left between the border of the shape or line and the text. This is similar to the settings for indentation and spacing for paragraphs.

The text anchor grid in the bottom right corner of the dialog is used to decide where to anchor the text. The **Full width** option determines if the anchoring should be applied to the full width of the shape.

Text Animation

Use the *Text Animation* tab to add special effects to the text. Choose between the four options on the list and, when applicable, the direction of the effect by picking one of the four arrow buttons to the right. The available effects are:

- **Blink:** the text will blink on the screen.
- **Scroll through:** the text will move into the shape and then out, following the chosen direction.
- **Scroll back and forth:** the text will move first in the chosen direction but will bounce back at the shape border.
- **Scroll in:** the text will scroll in towards the given direction, starting from the edge of the shape and stopping in the center.

The default is no animation.

The other properties that can be set are:

- **Start inside** option: when set, the animation will start from inside the shape.
- **Text visible when editing** option: Set this box to see the text while editing.
- **Animation cycles**: includes three further options to set the frequency of the animation, the increments between two positions of the animation, and finally, the delay before the animation starts.

To see some of the animations in action, you need to start the presentation. Press *F9* or select **Slide Show > Slide Show** from the main menu. To return to the edit mode, press *Esc*.

Formatting Connectors

Connectors are lines that join two shapes. Connectors always start from a **glue point** on the shape. Refer to Chapter 5 (Creating Graphic Objects) for a description of the usage of connectors.

Connector properties can be accessed and modified in two ways:

- Manual formatting: right-click on the connector line and select Connector in the pop-up menu.
- Style-based formatting: select one of the available graphics styles or create a new one.

Both methods open the Connector dialog, like that in Figure 32, where you can set the style of the connectors. Choose between Standard (the default), Line, Straight, and Curved connector. Whenever multiple connectors overlap, use the Line skew section of the dialog to distance the lines. It is possible to customize the distance between 4 different lines.

In the *Line spacing* section of the dialog, set the horizontal and vertical space between the connector and the object at each end of the connector.

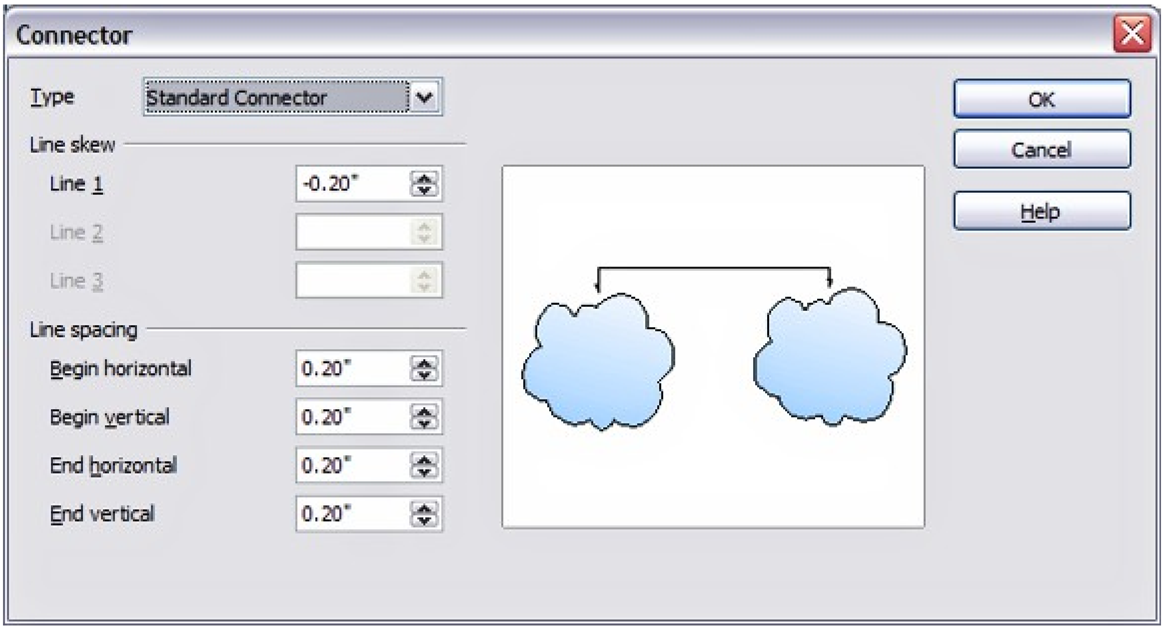


Figure 32: Setting up the connector properties

Working with Graphics Styles

To achieve consistency in the style across the slides of a presentation (or a portfolio of presentatios), or simply to apply the same formatting to a large number of objects, the best approach is to use graphics styles.

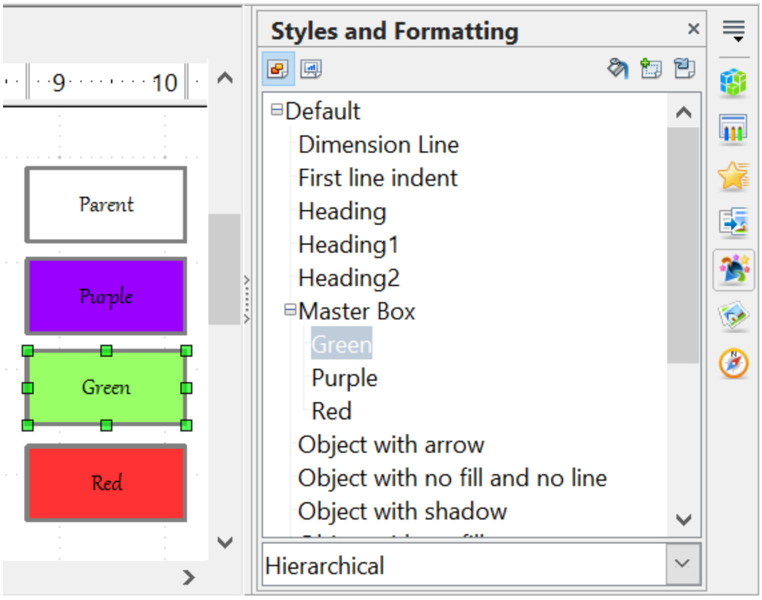
Graphics styles are the equivalent of paragraph styles for graphic objects. A graphics style groups all the formatting attributes that a graphic object can have and associates them with a name, making them quickly reusable. If a style is modified (for example, by changing the area transparency), the changes are automatically applied to all graphics using that style.

If you use Impress frequently, a library of well-defined graphics styles is an invaluable tool for speeding up the process of formatting your work according to your taste or any style guidelines you may need to follow (company colors, fonts, and so on).

Use the Styles and Formatting window to access styles you will need often. If the window is not visible, select its icon on the Sidebar, or press *F11*, or click the Styles and Formatting icon at the left-hand end of the formatting bar, or select **Format > Styles and Formatting** from the menu bar. If you use the windows outside of the Sidebar, press *F11* again when the dialog is no longer needed to maximize the workspace area.

Linked Graphic Styles

Graphic styles support inheritance; that is, a style can be linked to another (parent) style so that it inherits all the formatting settings of the parent. You can use this property to create "families" of styles.



For example, if you need multiple boxes that differ in color but are otherwise identically formatted, the best way to proceed is to define a generic style for the box, including borders, area fill, font, and so on, and a number of hierarchically dependent styles that differ only in the fill color attribute. If you need to change the font size or the border thickness later, it is sufficient to change the parent style. All the other styles (sometimes called children) will change accordingly.

Creating Graphics Styles

You can create a new graphics style in two ways:

- Using the Style dialog
- From a selection

Creating a New Graphic Style Using the Style Dialog

Choose the Graphics Styles icon  at the top of the Styles and Formatting window.

To link a new style with an existing style, as illustrated in Figure 33, first select that style, and then right-click and choose **New**. Otherwise, select *Default*, then right-click and choose **New**.

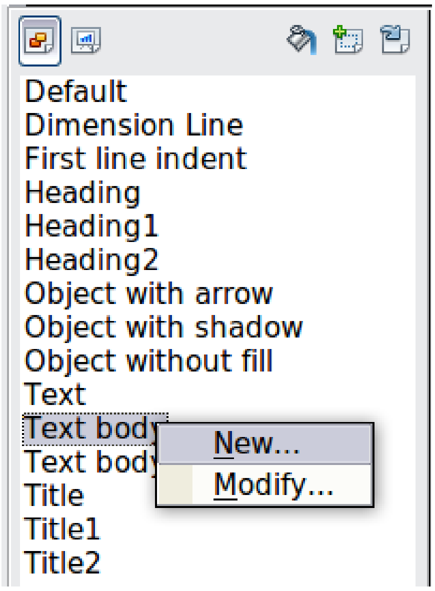


Figure 33: Linking a new style with an existing style

As discussed in the preceding section, when styles are linked, changing a property of the parent style will change it in all the linked styles. Sometimes this is exactly what you want; at other times, you do not want the changes to apply to all the linked styles. It pays to plan ahead.

Figure 34: Top of dialog to create a new graphics style

The Graphics style dialog shown in Figure 34 consists of 14 tabs (15 if Asian language support has been enabled), which may be grouped as follows:

- The Organizer tab contains a summary of the style and its hierarchical position.
- The Font, Font Effects, Indents & Spacing, Alignment, Tabs, and Asian typography tabs set the properties of the text inserted in a text box or in a graphic object.
- The Dimensioning tab is used to set the style of dimension lines.
- The Text, Text animation, Connector, Line, Area, Shadowing, and Transparency tabs determine the formatting of a graphic object and are discussed elsewhere in this chapter.

Note	In most cases, you will not need to configure the parameters of every tab. For example, to create a simple line style, you might use only 3 of the 15 tabs.

Creating a New Graphics Style from a Selection

You can create a new style from manually formatted text or graphics:

1) Select the item you want to save as a style. If the selected object is already styled, then the new style will be linked to such style.
2) In the Styles and Formatting window, click the **New Style from Selection** icon, highlighted in Figure 35.

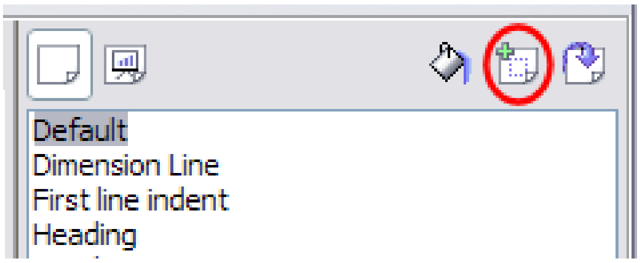


Figure 35: New style from selection

3) In the Create Style dialog, like that in Figure 36, type a name for the new style. The list shows the names of existing custom styles of the selected type. Click **OK** to save the new style.

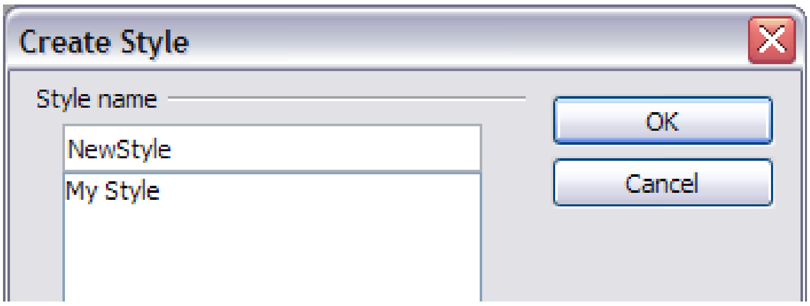


Figure 36: Naming a new style created from a selection

Modifying a Graphics Style

To change an existing style, right-click on it in the Styles and Formatting window and choose **Modify** from the pop-up menu.

The dialog for modifying a graphic style is the same as the one for creating a new graphic style.

Make the required changes to the style and then click **OK** to save them.

Updating a Graphic Style from a Selection

To update a style from a selection:

1) Select an item that has the format you want to adopt as a style.
2) In the Styles and Formatting window, select the style you want to update, and then click the **Update Style** icon shown in Figure 37.

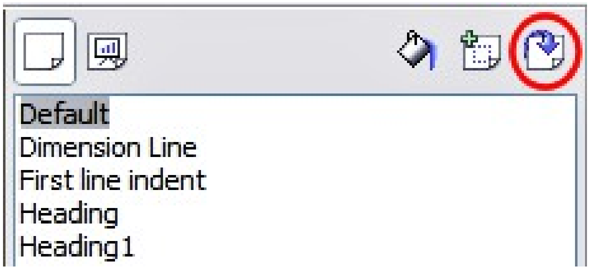


Figure 37. Updating a style from a selection

Tip	Any changes you make to a style are effective only in the document on which you are working. The changes do not go into any associated template. If you want the changes to apply to more than one document, you need to change the template (see Chapter 2).

Applying Graphics Styles

You can apply a graphics style in two ways, both of which start from the Styles and Formatting window. First, make sure that the graphic styles are shown, then do one of the following:

- Select the object to which you want to apply a graphic style and double-click on the name of the style you want to apply.
- Click the **Fill Format mode** icon  and then click on the name of the style you want to apply. When you move your cursor over the workspace, the mouse pointer changes to this icon. Position the moving icon on the graphic object to be styled and click the mouse button. This mode remains active until you turn it off, so you can apply the same style to several objects. To quit Fill Format mode, click the **Fill Format mode** icon again. When Fill Format mode is active, a right-click anywhere in the document undoes the last Fill Format action. Be careful not to accidentally right-click and thus undo actions you want to keep.

Tip	At the bottom of the Styles and Formatting window is a drop-down list. You can choose to show all styles or groups of styles such as applied styles or (in the case of graphics styles) custom styles.

Deleting Graphics Styles

You cannot delete any of the predefined styles, even if you are not using them.

You can delete any user-defined (custom) styles, but before doing so, you should make sure the styles are not in use. If an unwanted style is in use, replace it with a substitute style.

To delete unwanted styles, right-click on them (one at a time) in the Styles and Formatting window and click **Delete** on the pop-up menu. Click **Yes** in the message box that pops up.

Assigning Styles to Shortcut Keys

AOO provides a set of predefined keyboard shortcuts that allow you to apply styles while typing in a document quickly. You can redefine these shortcuts or define your own, as described in Appendix A (Keyboard Shortcuts).